

Experiment 3: Bright-field light microscope and mounting techniques

Principle:

The microscope is an important tool in the study of microorganisms as microorganisms are invisible to the naked eye. The **bright-field light microscope** magnifies images using two lens systems and visible light to directly illuminate specimens. The resulting image appears dark against a bright background. The two lens systems in a compound light microscope are the **ocular lens or eyepiece**, and the **objective lenses** located in the revolving nosepiece. Initial magnification occurs in the **objective lens**. There are three types of objective lenses: the **low-power, high-dry** and **oil immersion objective**. Magnification of microscope is the multiplication product of the powers of objective and ocular lenses.

Objective:

- Become familiar with the parts of light microscope
- Know proper use of the microscope
- Determine the total magnification by the compound microscope

Parts of a bright-field light microscope

A. Supporting stand includes:

- 1) **Base**- to hold the microscope and houses the illuminator or lamp
- 2) **Arm**- to support the optical system and house the fine adjustment
- 3) **Mechanical Stage**: a platform on which the object to be examined is placed. Often it is equipped with **stage clips**, A hole in the center of the stage, called the **Aperture**, to let light from the illuminator pass through.
- 4) **Body tube**- which supports **ocular lens (eye piece)** at top and a set of **objective lenses** attached to revolving **nose piece** at the lower end.

B. Optical System:

- 1) **Ocular lenses/Eye pieces**- is what you view through. It has magnifying power o.
- 2) **Set of objective lenses (4X, 10X, 40X, and 100X- oil immersion)** - Primary lenses magnify the specimen.
 - a. **The low-power: (10x)** or also called **16 mm objective**
 - b. **The high-dry: (40x)** or **4 mm objective**

c. The oil immersion: (100x) or 1.8 mm objective.

Immersion oil (which has the same **refractive index** as glass) is added on the space between the Oil immersion objective and the specimen to increase the **refractive index**.

- 3) **Condenser:** a lens focuses the light through a central opening in the stage on the object to be examined. It is fitted beneath the stage often in conjunction with an iris diaphragm
- 4) **Condenser Focus Knob** moves the condenser up or down to control the lighting focus.
- 5) **Iris Diaphragm:** controls the intensity of light entering the object through condenser
- 6) **Coarse and Fine focus knobs:** used for focusing to get sharp image. The objective can be moved away or closer to the object through **coarse** (for low power objective) and **fine adjustment knobs** (for high power and oil immersion objective)
- 7) **Illuminator:** the light source in the base of the microscope.

Proper use of the Microscope

Microscopes are very expensive pieces of scientific equipment and must be treated with care.

- 1) ***Moving and transporting the microscope.*** Always carry a microscope with two hands, one on the base and the other on the arm.
- 2) ***Adjusting the eyepiece.*** First find the proper distance between your eyes and the eyepieces by closing one eye and slowly moving your head toward that eyepiece until you see the complete field of view— about 1 inch away. Keep your head steady and both eyes in the same plane. Now open the other eye and gradually increase the distance between the eyepieces until it matches the distance between your eyes. At the correct distance, you will see one circular field of view with both eyes.
- 3) ***Positioning the slide.*** Place the slide specimen-side-up on the stage at the middle of the opening. Secure the slide between the slide clips.
- 4) ***Adjusting the illumination:***
 - a. Adjust the *amount of light coming through the condenser* using the *iris diaphragm lever*. For oil immersion objective and while viewing stained objects, the iris diaphragm should be set almost all the way open (maximum light). For low power objectives and while observing unstained objects, the iris should be set mostly closed (minimum light).
 - b. The *condenser* should be set all the way up (topmost position)

5) *Focusing from lower power to higher power:*

- a. Rotate the *yellow-striped 10X objective* until it locks into place
- a. Turn the *coarse focus* to raise the stage until it reaches an upward stop. Make sure to watch from the side to make sure this stop occurs. Do not allow the objective lens to hit the slide, while looking at the microscope from the side
- a. Look through the eyepieces and turn the *coarse focus* counterclockwise to lower the stage with the coarse adjustment until an image appears. Use the fine focus knob to desired sharp image. Adjust the light for optimum contrast using the iris diaphragm.
- b. If higher magnification is desired, simply rotate the *blue-striped 40X objective* into place and the specimen should still be in focus. (Minor adjustments in fine focus and light contrast may be needed.)
- c. For magnification with *100X or oil immersion*, rotate the *bluestriped 40X objective* slightly out of position and place a *drop of immersion oil* on the center of the slide. Now rotate the *white-striped 100X oil immersion objective* into place. Again, the specimen should remain in focus, although minor adjustments in fine focus and light contrast may be needed.

- 6) ***Cleaning the microscope and returning to place.*** After finished, Place the low-power objective in line with the ocular; lower the stage to its lowest position. Clean the lenses with lens paper only and lens cleaner. *Remove any oil from the oil immersion lens.* Cover and return the microscope to its proper storage place.

Discussion questions

1. Which objective focuses closest to the object?
2. What would occur if you use water instead of immersion oil?

Experiment4: Preparation of bacterial smear and staining

Principle:

Visualization of microorganisms in the living state is very difficult since they are small, transparent, and motile when suspended in fluids. Bacterial specimens are often stained to increase their **contrast** so that they can be more clearly seen in the bright-field microscope. Dyes are organic compounds that adhere to a cell and giving color to a cell. The dyes are classified into two main types: basic dyes (positively charged) and acidic dyes (negatively charged). Examples of basic dyes include methylene blue, crystal violet, safranin and carbol fuchsin. Acidic dyes include such as nigrosin, India ink, and picric acid. Because cell surfaces tend to be negatively charged, basic dyes will combine with high affinity to the cell walls and hence are excellent general-purpose stains. Whereas, a negative dye will be repulsed by the cell wall and not directly stain the cell.

Staining procedures are classified as simple stain and differential stain. A **Simple stain** will stain all cells with one color, A **Differential stain** uses a sequence of two contrasting stains that allows differentiating between groups of bacteria, or specific structures in a bacterium. Common examples are the **Gram stain and Acid-fast stain**.

A **smear** of is prepared on a glass slide and **heat-fixed** before a specific staining process. A **smear** is preparation of cell suspensions spread and dried on a glass slide. **Fixation** is accomplished by passing the air-dried smear through the top of Bunsen burner flame 2 to 3 times. Fixation denatures the bacterial enzymes and prevents autolysis and ensures bacterial adherence to microscopic slide

Objectives:

- Learn the proper preparation of heat-fixed bacterial smears
- Become familiar with simple staining procedure
- Observe and examine morphology and arrangements of microbial cells

Materials and Equipments:

Specimen (broth culture)	Staining racks	Microscope slides
Inoculating loop	Tap water	Bunsen burner
Methylene blue	Immersion oil	Lens paper and lens cleaner

Procedure:

A. Preparing a bacterial smear

1. Label a clean microscope slide with a marker pen; this is also a useful reminder of which side of the slide is being used.
2. Flame a wire loop to ensure that no culture accidentally remains from a previous operation.
3. Place one loop of tap water on the center of the slide. Flame and cool the loop and touch a small part of a colony from a plate.
4. Make a suspension of the culture in the water on the slide and evenly but gently spread it into a thin film (approximately 1cm²)
5. Flame the loop. Air-Dry the smear, and “fix” it by quickly passing through the flame 2 to 3 times. The slide should not be burning hot. This is called a **heat-fixed smear**; it should be visible to the naked eye as a whitish area.

B. Simple stain

1. Prepare a heat-fixed film as described in the previous section
2. Put the slide with the smear uppermost on a staining rack over a sink or staining tray.
3. Cover the film thoroughly with methylene blue solution for 1 to 2 min.
4. Tilt the slide to allow the excess stain to run off into the staining tub, and wash gently with the water
5. Allow the slide to air dry, or gently blot dry with filter paper
6. Examine the stained slide with your microscope using the 40X and oil immersion objectives
7. When finished, dispose of slides into discard jar

Discussion Questions

1. Draw the morphological forms observed under the microscope
2. Do all bacteria stain with simple stains?
3. What would happen if you apply too much heat while heat fixing the smear?
4. What is the advantage of simple stain?

Experiment 5: Gram staining technique

Principle

The gram stain is named for Christian Gram who developed it in 1884. The Gram stain separates bacteria into two groups : (1) Gram-positive bacteria that retain the Crystal violet and (2) Gram-negative bacteria that take the color of the counterstain(usually Safranin). This reaction is due to differences in the composition of the cells. Gram positive sells primarily consists of a thick peptidoglycan layer (90%). whereas Whereas, **Gram negative cell wall** is multilayered consisting of a thin layer of peptidoglycan (10%), and additional **outer membrane of Lipopolysaccharides**.

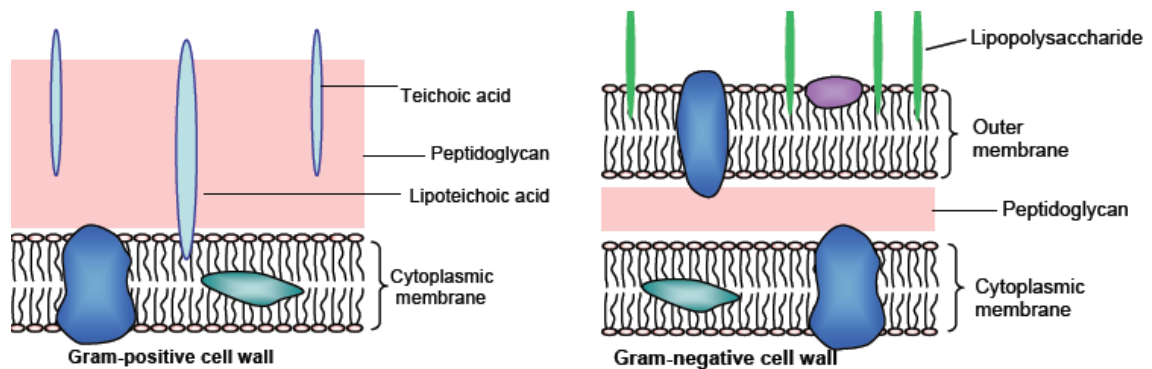


Figure 1 Cell wall structure in gram positive and gram negative bacteria

In the gram staining procedure, the **primary stain** is crystal violet, and all cells take up the crystal violet dye. The second step (Gram's Iodine, a mordant) stabilizes the Crystal violet into the peptidoglycan layer of the cell wall (forms crystal violet-iodine complex and locks it into peptidoglycan). The third step (alcohol decolorization) dissolves lipids in the outer membrane of Gram-negative bacteria and removes the Crystal violet from the peptidoglycan layer. In contrast, the Crystal violet (since it is extensively entrapped in the thick peptidoglycan) is relatively inaccessible in Gram-positive microorganisms and cannot readily be removed by alcohol. After the alcohol step, only the colorless Gram-negative microorganisms can accept the Safranin (counterstain). Thus, at the end, Gram-positive cells are purple/violet and Gram-negative cells are pink/red.

The decolonization step is the most problematic part of the procedure. The amount of alcohol treatment (the differential stage) must be judged carefully because over treatment washes the CVI complex from Gram-positive bacteria and they will appear to be Gram-negative. Thick slide preparations and excessive heat fixing that distorts the bacteria affect the decolorization. It is preferable to use fresh cultures for the Gram stain. Old cultures may stain "Gram-variable" (a mix of purple and pink) because they decolorize easily.

Objectives:

- Perform a satisfactory Gram stain
- Know the dyes and reagents, their functions and steps in Gram staining
- Differentiate a mixture of bacteria into gram positive and gram negative cells

Materials and Equipments

18 to 24 hour tryptic soy broth cultures

Gram's Crystal violet

Gram's iodine

Tap water

95% ethanol or isopropanol-acetone (3:1 v/v) or ethanol-acetone (1:1 v/v)

Safranin

Inoculating loop

Bunsen burner

Bibulous paper/filter paper

Staining rack

Microscope slides

Procedure

1. Prepare a heat-fixed smear as described in the previous section
2. place the slide with the fixed smear uppermost on a staining rack
3. Thoroughly cover the smear with crystal violet solution and leave for 1 minute.
4. Pour off the stain, and Wash with tap water.
5. Cover the smear with Gram's iodine for 1 min. Wash with tap water
6. Hold the slide at a 45° angle over the sink, decolorize with 95% ethanol for 15 to 30 seconds. Add the decolorizer drop by drop until the washings are pale violet. Alternatively a mixture of isopropanol-acetone can be used for 30 to 60 seconds.
7. Rinse immediately with tap water.
8. Cover the smear with the safranin for 1 minute. Wash with tap water.
9. Blot-dry, examine under oil immersion. G-positive cell appear purple/violet and G-negative cells stain pink to red.

Discussion Questions

1. Why do the old Gram-positive cultures stain Gram negative?
2. What do you conclude if you see both rods and cocci in Gram stained field?
3. What part of the bacterial cell is most involved with Gram staining, and why?